GAO Ziming

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Research Interests

Human-computer interaction; XR interaction; human-AI interaction; context-aware assistance for physical activities.

Educational Background

The University of Hong Kong 09/2026-present

MSc(Eng) in Innovative Design and Technology; expected completion: 07/2027

Beihang University (BUAA) 09/2021-06/2025

BEng in Industrial Design

- CGPA: 3.37/4.0; Major GPA: 3.67/4.0; Last-two-year GPA: 3.63/4.0
- BUAA scholarships: Top Prize of Encouragement Scholarship (top 5%) & Second Prize of Merit Scholarship (2023/24); Second Prize of Encouragement Scholarship (2022/23)

Research Experience

Situated Reality Lab, The University of Hong Kong 09/2026-present

MSc Dissertation Research & Research Assistant · Supervisor: Prof. Xiang Li

XR Microgestures during Bimanual Object Handling

Early-stage research; target venue: UIST 2027.

- Investigating small command gestures while both hands hold or manipulate an object, with minimal disruption to the primary task.
- Planning gesture elicitation and recognition studies; comparing rigid-panel and fabric inspection scenarios and evaluating task stability.

IDEA Lab, Zhejiang University 03/2026-present

Research Intern; Focus: Human-AI interaction and context-aware assistance for physical activities.

- Presented the group's research on contextual AI assistance at the IDEA Lab Summer Meeting, Zhejiang University, on 31 July 2026.

Physical-Knowledge Grounding

Manuscript submitted to CHI; second author.

- Conducted questionnaires and interviews on adapting online tutorials to users' materials, tools, and environments.
- Synthesized findings into design goals; created paper figures and supported user-study design and execution.

Situation-Bound Intent

Manuscript submitted to ACM TOCHI; fourth author.

- Designed modular wearable hardware for situation-aware prospective memory assistance.
- Created user-study result figures and refined other manuscript visuals.

School of Mechanical Engineering & Automation, BUAA 09/2025-03/2026

Part-time Research Assistant, High-temperature and High-pressure Flexible Forming Laboratory

- Performed Abaqus simulations of component forming; analyzed and visualized simulation results.

Publication

Gao, Z., & Xu, W. (2026). Product innovation design based on the sail-shaped structural characteristics of Velella. Journal of Northwest Normal University (Natural Science), 62(4), 92–100. In Chinese. DOI: 10.16783/j.cnki.nwnuz.2026.04.009.

Selected Projects and Awards

VelellaNexus — Undergraduate Graduation Project

- Developed a biomimetic floating platform through morphology research, experiments, simulation, and prototyping.
- NCDA 2025: National Second Prize and Beijing First Prize;
  First Prize, 13th Beijing Universities Industrial Design Competition.

Skills and Teaching Experience

- Research: Formative studies, questionnaire and interview synthesis, research visualization.
- Tools: Figma, Arduino, Processing, Rhino, SolidWorks, KeyShot, Illustrator and Photoshop.
- Languages: Mandarin Chinese (native); English (IELTS 6.5); Japanese (JLPT N2).
- Teaching Assistant, BUAA: Product Design (Intelligent Interaction), Interactive Programming Techniques, and Illustration.